REPORT PROJECT 2

Sudoku problem :

Task 1 : How do you represent a solution to the problem ?

A possible solution to the sudoku problem is as follows:

$$X = \{ [X_{1,1}, X_{1,2}, \dots, X_{1,9}],$$

$$[X_{2,1} \dots, X_{2,9}]$$

$$\dots$$

$$[X_{9,1} \dots, X_{9,9}] \}$$

with

$$X_{i,j} \in [\![1, 9]\!], \ \forall \ i, j \in [\![1, 9]\!]$$

where $X_{i,j}$ represents the number in the i-th line and the j-th column.

$X_{1,1}$	$X_{1,2}$	$X_{1,3}$	$X_{1,4}$	$X_{1,5}$	$X_{1,6}$	$X_{1,7}$	$X_{1,8}$	$X_{1,9}$
$X_{2,1}$	$X_{2,2}$	$X_{2,3}$	$X_{2,4}$	$X_{2,5}$	$X_{2,6}$	$X_{2,7}$	$X_{2,8}$	$X_{2,9}$
$X_{3,1}$	$X_{3,2}$	$X_{3,3}$	$X_{3,4}$	$X_{3,5}$	$X_{3,6}$	$X_{3,7}$	$X_{3,8}$	$X_{3,9}$
$X_{4,1}$	$X_{4,2}$	$X_{4,3}$	$X_{4,4}$	$X_{4,5}$	$X_{4,6}$	$X_{4,7}$	$X_{4,8}$	$X_{4,9}$
$X_{5,1}$	$X_{5,2}$	$X_{5,3}$	$X_{5,4}$	$X_{5,5}$	$X_{5,6}$	$X_{5,7}$	$X_{5,8}$	$X_{5,9}$
$X_{6,1}$	$X_{6,2}$	$X_{6,3}$	$X_{6,4}$	$X_{6,5}$	$X_{6,6}$	$X_{6,7}$	$X_{6,8}$	$X_{6,9}$
$X_{7,1}$	$X_{7,2}$	$X_{7,3}$	$X_{7,4}$	$X_{7,5}$	$X_{7,6}$	$X_{7,7}$	$X_{7,8}$	$X_{7,9}$
$X_{8,1}$	$X_{8,2}$	$X_{8,3}$	$X_{8,4}$	$X_{8,5}$	$X_{8,6}$	$X_{8,7}$	$X_{8,8}$	$X_{8,9}$
$X_{9,1}$	$X_{9,2}$	$X_{9,3}$	$X_{9,4}$	$X_{9,5}$	$X_{9,6}$	$X_{9,7}$	$X_{9,8}$	$X_{9,9}$

Task 2 : What is the equation for the restriction(s) of the problem ?

The problem has the following constraint :

$$C = \{ X_{i,m} \neq X_{i,n} , \forall\, i,\ m ,\ n \ \ and\, m \neq n\}$$

$$\cup \{ X_{m,j} \neq X_{n,j} , \forall\, j,\ m,\ n\, and\, m \neq n\}$$

$$\cup \{ X_{(x//3)\times 3+i,(y-1//3)\times 3+j} \neq X_{(x-1//3)\times 3+m,(y//3)\times 3+n}$$

$$\forall x, y \in [\![0, 8]\!],\ i, j, m, n \in [\![1, 3]\!]\ and\, m \neq n,\ \ i \neq j \ \}$$

Task 3 : What is considered a state?

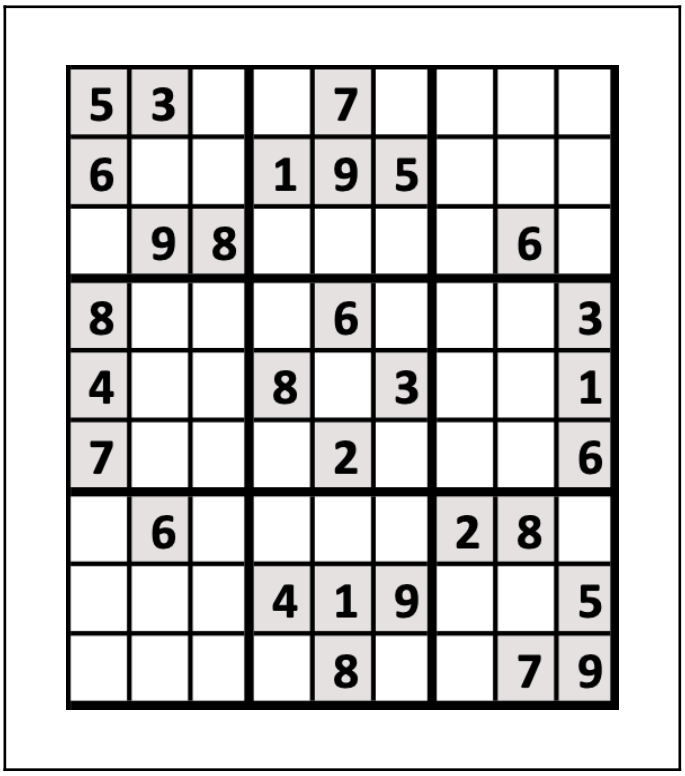
A state can be considered as a 9x9 Sudoku grid where every cell can be either empty or with a number from 1 to 9, while still following the constraints for all the not empty cells.

For example :

5	3		6	7	8			2
6			1	9	5			8
1	9	8	3	4	2	5	6	7
8				6				3
4		6	8	5	3			1
7		3		2		8		6
	6			3		2	8	4
	8		4	1	9			5
				8			7	9

Task 4 : Which is the initial state?

A sudoku grid before solving, for example :



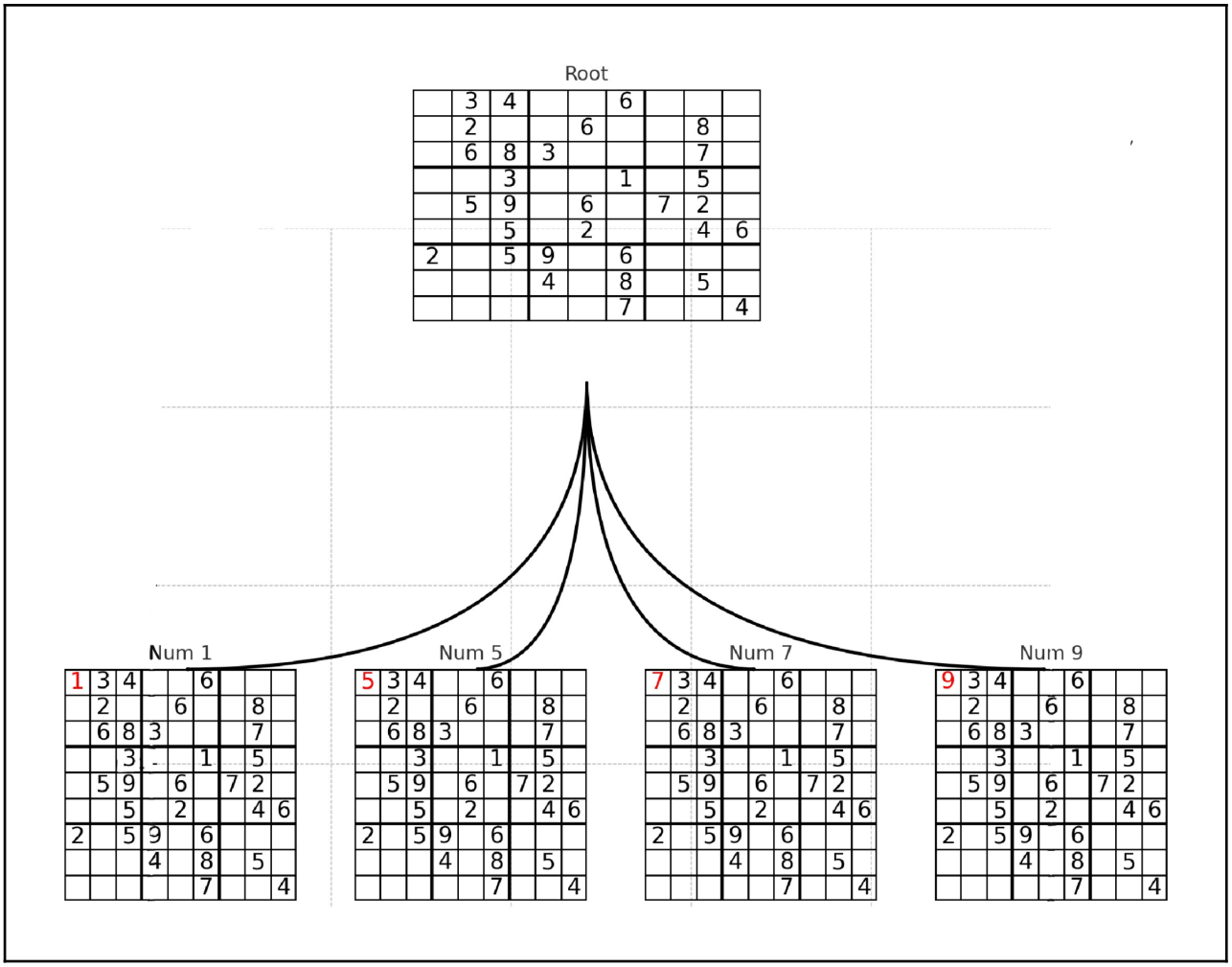
Task 5 : Which is/are the possible action(s)?

A possible action is replacing an empty cell by a number from 1 to 9, while respecting the constraints.

Task 6 : What is the maximum branching factor (b) ?

For every empty cell, there are a number of possible “replacement numbers” for that spot depending on the constraints. For every possibility, a branch or child is created with that “replacement number”. In the worst case, the branching factor is 9 (maximum), if all the numbers (1 to 9) are valid in an empty cell. Generally, the more we progress in filling the sudoku, the less branches are created since the constraints are stricter.

For example :



Task 7 : What is the maximum depth (m) ?

The maximum depth is equal to the initial total number of empty cells we had to fill, since for every empty cell, by choosing any “replacement number”, we extend the tree’s depth by one. In a very unrealistic worst case, where the initial sudoku is empty, the depth would be 9x9 = 81.